AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII EXAMINATION

APRIL/ MAY 2019

Mathematics Paper II

Time: 2 hours 10 minutes Marks: 65

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INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's Signature**

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2. There are EIGHT questions. Answer ALL questions. Choices are specified inside the paper.

3. When answering the questions:

   Read each question carefully.
   Use black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a scientific calculator if you wish.

Q.1. (Total 5 Marks)

Find the following limits.

a. $\lim_{x\to 3} \frac{\left(x^2-9\right)^2}{2x^2-12x+18}$ (3 Marks)

b. $\lim_{x\to 0}\left(\frac{e^{e^x}-e^x}{e^{e^x}+e^x}\right)$ (2 Marks)

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.2.)

Q.2. (Total 12 Marks)

a.

i. If $y = e^{x+\tan x}$, then show that $\frac{dy}{dx} = 2y + y\tan^2 x$. (3 Marks)

ii. For $y = \left(\sqrt{3x-2}\right)\left(\sqrt{3x+2}\right)$, show that $\frac{dy}{dx} = \frac{9x}{\sqrt{9x^2-4}}$. (3 Marks)

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.2.)

b. For $y = \ln\left(\frac{\sqrt{x\cos x}}{\sin x}\right)$,

i. find $\frac{d^2y}{dx^2}$ (5 Marks)

ii. Hence, find the value of $\frac{d^2y}{dx^2}$ at $x = \frac{\pi}{4}$ (1 Mark)

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c. A water tank is cylindrical in shape. If the volume of the tank is 64 m^3, then what would be the radius and height of the water tank to have the least surface area? Write you answer to two decimal places. (6 Marks)

(**Note**: The volume of a cylinder is $V = \pi r^2 h$ and surface area of a cylinder is $S = 2\pi r^2 + 2\pi rh$.

Where *r* is the radius of the cylinder and *h* is the height of a cylinder.)

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.3.)

Q.3. (Total 12 Marks)

a.

i. Convert $\frac{3x}{(x^2-4)(x^2+4)}$ into its partial fractions. (4 Marks)

ii. Hence, find $\int \frac{3x}{(x^2-4)(x^2+4)}\,dx$ (2 Marks)

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.3.)

b. Evaluate the following indefinite integrals.

i. $\int \left(4x^4 - 12x^2 + 9\right)^{\frac{3}{2}} x\,dx$ (3 Marks)

ii. $\int \frac{x}{(x+b)^2}\,dx$ (3 Marks)

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.3.)

c. Evaluate the following definite integrals by showing all the necessary steps. Give you answer in terms of *a*. (6 Marks)

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{dx}{a + a\cos x}$$

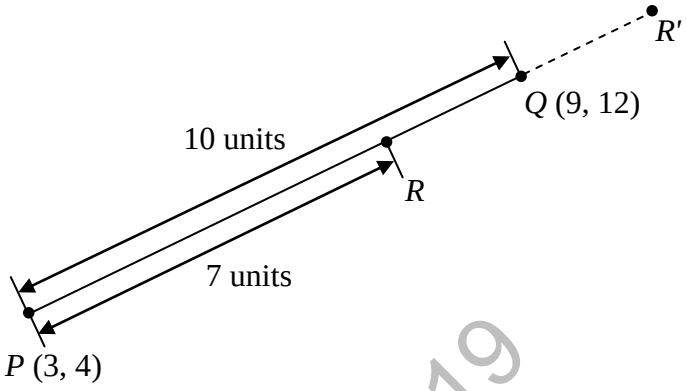
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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.4.)

Q.4. (Total 10 Marks)

a. In the given diagram, the length of *PQ* and *PR* are 10 units and 7 units respectively.

i. Find the coordinates of *R*. (3 Marks)



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ii. If *PQ* is produced to a point R' such that $RQ = QR'$, then find the coordinates of R'. (2 Marks)

(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.4.)

b. Find the equation of the line passing through the point of intersection of the lines $x+3y=6$ and $3x+2y=11$, which

i. is perpendicular to the line $5x-y=3$. (3 Marks)

ii. passes through the point (2, 3). (2 Marks)

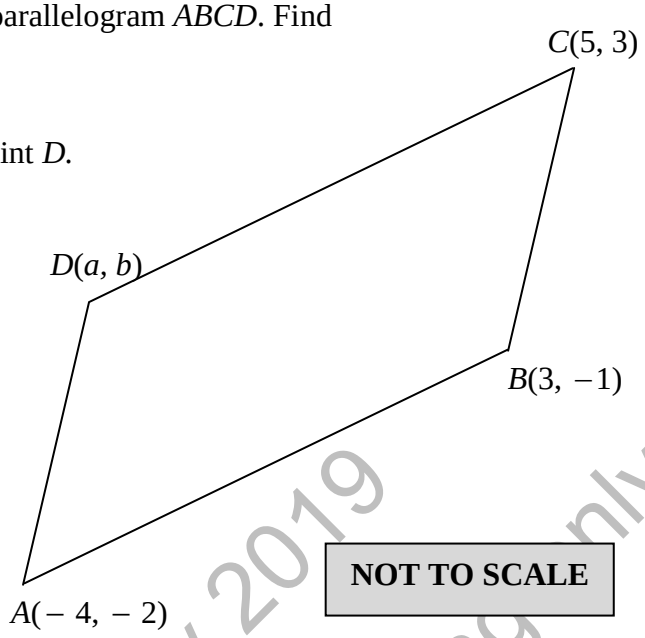
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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.4.)

c. The given figure shows a parallelogram *ABCD*. Find

i. the angle *ABC*. (3 Marks)

ii. the coordinates of point *D*. (2 Marks)



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Q.5. (Total 6 Marks)

A burger shop sells chicken and beef burgers. The profit on chicken burger and beef burger is Rs 12 and Rs 10 respectively.

Due to existing cooking facilities, it cannot cook

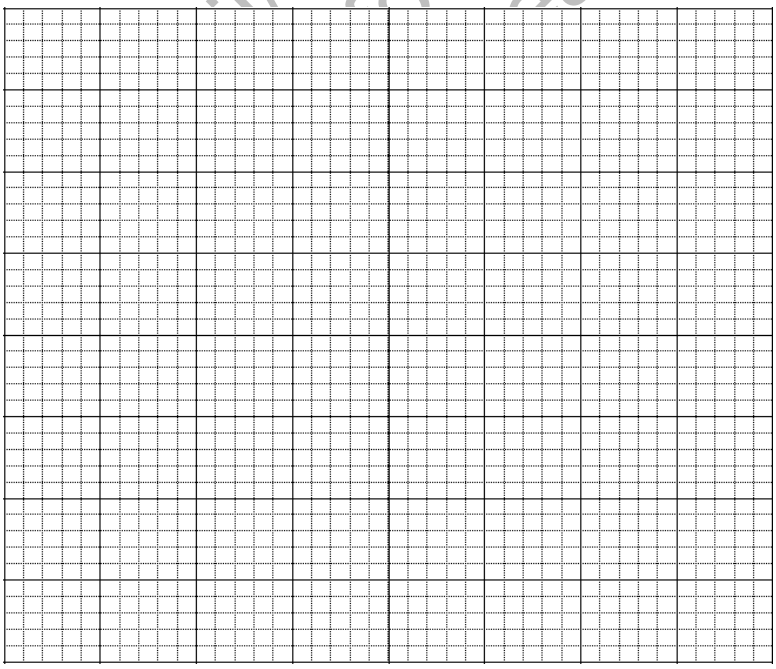
- more than 200 chicken burgers.
- more than 250 beef burgers.
- and sell more than 400 burgers altogether.

For the given linear programming problem,

i. state the constraints (1 Mark)

ii. state the profit function (1 Mark)

iii. draw constraints on the given graph to find feasible region and corner points. (3 Marks)



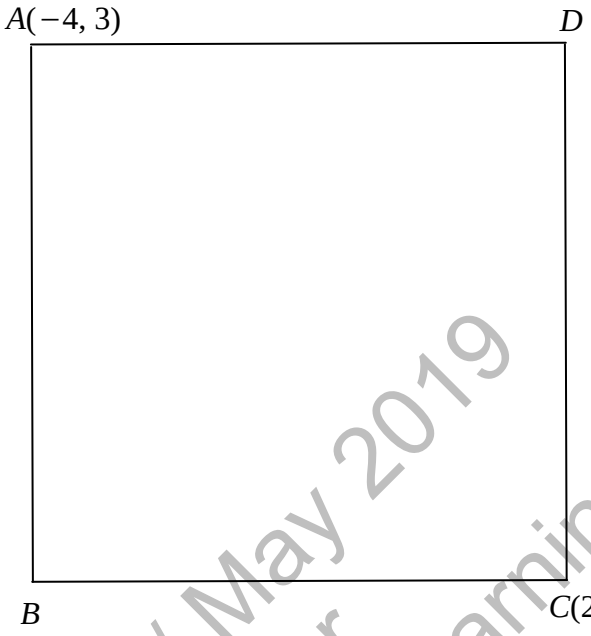
iv. find the maximum profit. (1 Mark)

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(ATTEMPT EITHER PART a OR PART b OF Q.6.)

Q.6. (Total 6 Marks)

a. In the diagram, a square *ABCD* is given. Find the equation and radius of the circle passing through the vertices of the given square. (6 Marks)



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(ATTEMPT EITHER PART a OR PART b OF Q.6.)

b.

i. Find the equations of tangent and normal to the circle $7x^2 + 7y^2 - 28x + 42y - 84 = 0$ at point $(-3, -3)$. (4 Marks)

ii. If $y = bx$ is a tangent to the circle $x^2 + y^2 = a^2$, then show that $4a^2(1 + b^2) = 0$. (2 Marks)

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.7.)

Q.7. (Total 10 Marks)

a. Find the vertices, co-vertices and eccentricity of the ellipse represented by (5 Marks)

$4x^2 + 49y^2 = 196$

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.7.)

b. An ellipse with centre $(0,0)$ has vertices $(\pm 5,0)$ and foci $(\pm 4,0)$. Find

i. the equation of the ellipse. (2 Marks)

ii. the eccentricity. (1 Mark)

iii. the equation of directrices. (1 Mark)

iv. the length of minor axis. (1 Mark)

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(ATTEMPT ANY TWO PARTS FROM a, b AND c OF Q.7.)

c. The vertices of a hyperbola are $(15, 1)$ and $(-1, 1)$. If the length of its conjugate axis is 12 units, then find the equation of the hyperbola in standard form. (5 Marks)

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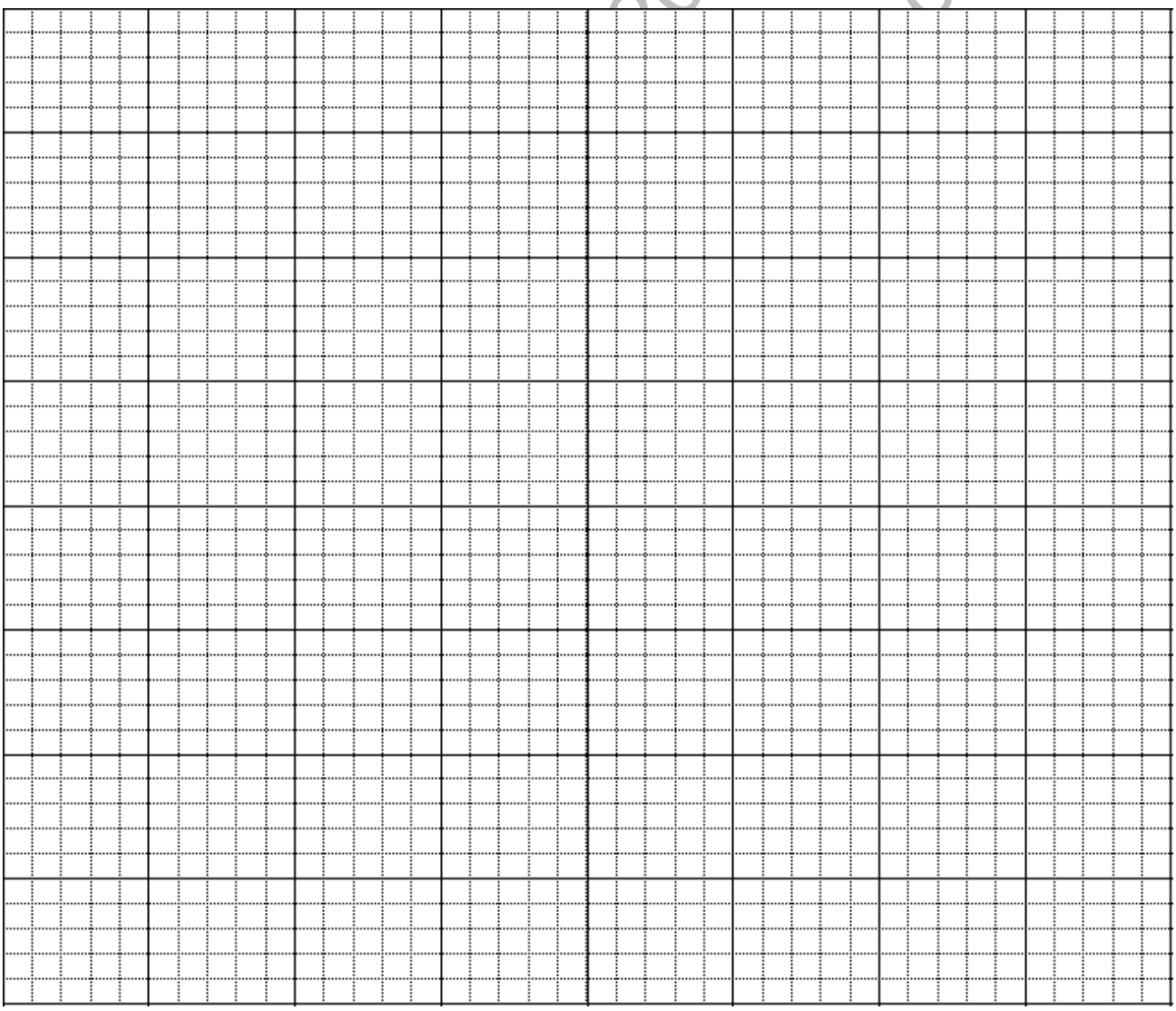
(ATTEMPT EITHER PART a OR PART b OF Q.8.)

Q.8. (Total 4 Marks)

a. A triangle *ABC* located on *xy*–coordinate axes has vertices *A* (0, 3), *B* (– 3, 0) and *C* (3, – 3). The *x*–axis is translated through a distance of – 3 units while there is no change in the *y*-axis.

i. Find the new vertices of the triangle *ABC* with respect to the translated axis. (3 Marks)

ii. Using the given graph paper, draw the triangle *ABC* with its new coordinates. (1 Mark)



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(ATTEMPT EITHER PART a OR PART b OF Q.8.)

b. The xy – coordinate axes are rotated about the origin through an angle of $60°$. The coordinates of a point A are (7, 11) with respect to xy – coordinate axis. Find coordinates of A with respect to rotated axes. (4 Marks)

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END OF PAPER

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